



Dao Thanh Nguyen

Data Engineer

SUMMARY

Data Engineer with 2+ years of professional experience specializing in big data systems, real-time streaming pipelines, and cloud data architecture. Ranked Top 2 in Data Engineering at Viettel Digital Talent 2024 program. Proven ability to design and optimize high-throughput ETL pipelines processing up to 5 million messages per second. Experienced in both on-premises big data infrastructure and cloud platforms (Azure). Strong background in Python, Java, and distributed computing frameworks, including Apache Flink, Spark, and Kafka.

Specialties

- Strong experience designing and implementing real-time streaming data pipelines using Apache Spark, Flink and Kafka at production scale (up to 5M messages/second).
- Strong experience building batch ETL pipelines using Apache Spark, optimizing for performance and data quality.
- Experience with cloud data platforms including Microsoft Azure (Event Hub, Stream Analytics, Databricks, Data Lake Storage Gen2, Azure Machine learning, Power BI).
- Strong experience with big data infrastructure including HDFS, Apache Iceberg and Kubernetes deployment.
- Experience debugging and resolving complex performance bottlenecks in distributed data systems (data skew, state management, checkpoint optimization).
- Experience with data lakehouse architecture using Delta Lake, Apache Iceberg, and Medallion (Bronze/Silver/Gold) patterns.
- Experience with Iot data pipelines - Azure Event Hub, Stream Analytics, InfluxDB for real-time sensor data processing.
- Experience applying machine learning models for real-time prediction in production research systems.
- Professional English proficiency — technical documentation, written communication, and international project collaboration.
- Ability to work independently and deliver end-to-end solutions with minimal supervision.

Industry Experience

- Telecommunications: VoLTE quality monitoring, 2G/3G/4G network data processing, network optimization analytics.
- Research & IoT: CO2 microalgae capture pilot monitoring, real-time operational parameter prediction.

TECHNICAL STACKS

TECHNICAL EXPERTISE & SKILLS

Proficiency Description

1-Basic knowledge: Basic knowledge, needs much guidance and support in using the competence

2-Limited experience: Able to work in the area, needs some guidance in using the competence

3-Practical application: Good knowledge and skills, works independently, has sound experience in using the competence.

4-Applied theory: Advanced knowledge and skills, provides guidance and direction to others

5-Recognized authority: Exceptional knowledge and skills, perceived as an authority in this competence area, is innovative in using the competence.

Technology skills

Skill	Proficiency	Experience	Last used	Notes
Programming Language				
Java	4-Applied theory	2y	2026	
Python	4-Applied theory	3y	2026	
SQL	4-Applied theory	3y	2026	
Streaming & Batch Processing				
Apache Flink	4-Applied theory	1.5y	2026	RocksDB, checkpoint, watermark tuning
Apache Spark	4-Applied theory	2y	2026	Salt key optimization, skew handling
Apache Kafka	4-Applied theory	2y	2026	Avro, Protobuf
Cloud Platforms				
Microsoft Azure	3-Practical application	1.5y	2024	
Azure Databricks	3-Practical application	1y	2024	

Data Storage & Lakehouse				
HDFS	4-Applied theory	2y	2026	
Apache Iceberg	3-Practical application	1y	2026	
PostgreSQL	2-Limited experience	1y	2026	
ClickHouse	2-Limited experience	1y	2025	
Infrastructure & Devops				
Kubernetes	3-Practical application	1y	2026	
Apache Airflow	3-Practical application	1y	2026	
Git / GitLab	3-Practical application	2y	2026	

EDUCATION

Bachelor of Data Science - VNUHCM University of Science, Vietnam

Oct 2020 – Aug 2024 | GPA: 3.47/4.0

- Thesis scored 10/10 — supervised by Dr. Trung Trinh (Sintef, Norway) & Dr. Ha Hoang Van: Development of YOLO machine learning model and real-time streaming operational parameters of CO2 microalgae capture pilot. Implemented YOLOv8-based gas bubble detection system with real-time video stream processing.

CERTIFICATES, HONORS AND AWARDS

- Viettel Digital Talent 2024 — Top 50 Ambassador across 9 fields, Ranked Top 2 in Data Engineering (Stream Processing — Apache Flink, Spark & Kafka)
- Microsoft Azure Data Fundamentals — DP-900 Exam Prep (Coursera)
- TOEIC Listening & Reading: 740

WORKING EXPERIENCE HIGHLIGHTS

Viettel Networks — Big Data Engineer

Mar 2024 – Apr 2026 | Vietnam

Project: VoLTEMon (2024 – 2026)

High-performance ETL batch & streaming pipeline for VoLTE network data. Transforms and enriches raw data, loading processed results into the Data Lakehouse for AI Engineers and Data Analysts.

Project size: 2 members.

Roles: Data Engineer - ETL Pipeline Development & Infrastructure

- Designed and developed batch and streaming data pipelines transforming raw VoLTE data into analytics-ready features.
- Implemented Apache Iceberg table format for ACID-compliant data lakehouse with time-travel capabilities.
- Deployed and managed pipeline infrastructure on Kubernetes with Argo CD for CI/CD.
- Orchestrated batch workflows using Apache Airflow.

Technologies: *Apache Flink, Apache Kafka, Apache Iceberg, Apache Spark, HDFS, MinIO, Parquet, Kubernetes, Airflow, Argo, Java, Scala, JDBC, GitLab*

Project: DATAMON (2024)

High-volume streaming big data system processing 80–150 million 3G & 4G network messages per minute with complex business logic. Solo ownership of debugging, optimization, and stabilization.

Project size: 2 members.

Roles: Data Engineer — Debugging & Optimization

- Identified and resolved critical data skew issue causing job failures and data quality degradation from 80% to 30%.
- Implemented salt key technique to redistribute skewed partition data, eliminating task timeouts and job crashes.
- Analyzed execution bottlenecks using Spark Web UI — stage timelines, task durations, shuffle metrics.
- Restored data quality to 85% and reduced batch processing time from 2–4 hours back to 3–4 minutes.

Technologies: *Apache Spark Streaming, Apache Kafka, HDFS, Java, YARN, Ambari*

Sintef Norway Research Support — Remote Data Engineer (Part-time)

Jul 2022 – Mar 2024 | Remote (Vietnam → Norway)

Project: Real-Time & Batch Data Pipeline - Azure Cloud (2022 - 2024)

End-to-end Azure cloud data platform for a CO2 microalgae capture research pilot in Norway. Supports real-time parameter monitoring, ML-based prediction, and interactive data exploration.

Project size: 2 members (1 Project Manager, 1 Data Engineer).

Roles: Data Engineer

- Designed and implemented real-time pipeline: Event Hub → Stream Analytics → Azure ML (trained models) → ADLS Gen2 → Power BI streaming dashboard.
- Built batch pipeline: Azure App Service (Flask) → Blob Storage → Azure Function App (trigger) → Databricks (ETL + ML prediction) → SQL Server → Power BI.
- Developed Streamlit application with login, ML prediction interface, and Azure SQL Server backend for researcher use.
- Benchmarked and deployed supervised and regression ML models for operational parameter prediction.

Technologies: *Azure Event Hub, Stream Analytics, Databricks, Azure ML Studio, ADLS Gen2, Azure Blob Storage, Azure Function App, Azure App Service, Azure SQL Server, Azure Active Directory, Key Vault, Power BI, Flask, Streamlit, Python*